**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**International General Certificate of Secondary Education**

# MARK SCHEME for the October/November 2013 series

## 0620 CHEMISTRY

**0620/23** Paper 2 (Core Theory), maximum raw mark 80

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**1** **(a)** **(i)** aluminium [1]

**(ii)** calcium and iron [1]

**(iii)** lithium [1]

**(iv)** silver [1]

**(v)** aluminium [1]

**(b)** Any 2 of: [2]

reacts with acids
rusts / reacts with water and oxygen / reacts with water and air
reacts with steam
reacts with oxygen
reacts with chlorine
acts as a catalyst
any other suitable e.g. reacts with nitrates of less reactive metal

**(c)** calcium oxide added / lime added [1]
oxygen / air (blown into molten iron) [1]

**[Total: 9]**

**2** **(a)** Any five of: [5]

nucleus in centre of atom
protons and neutrons in nucleus / protons and neutrons in centre of atom
electrons outside the nucleus / idea of electrons in shells outside the centre of atom
2 protons
2 electrons
2 neutrons (in commonest isotope)
protons positively charged
electrons negatively charged
neutrons have no charge

**(b)** airships / blimps / balloons / diving / lasers / any other suitable [1]

**(c)** 223 [2]
Xe =131, O =16, F = 19 (for 1 mark)

**(d)** **(i)** (room temperature): gas [1]
(-200 °C): liquid [1]

**(ii)** has two atoms [1]
**IGNORE**: $F_2$ / $Cl_2$ (unqualified) / reference to same atoms or different atoms

**[Total: 11]**


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**3** **(a)** 2 electrons in outer shell [1]
8 electrons in middle two shells [1]

**(b)** calcium chloride [1]

**(c)** **(i)** 27 $cm^3$ [1]

**(ii)** lower initial gradient [1]
ends up at same volume of gas [1]

**(iii)** temperature: goes faster / increases [1]
hydrochloric acid: goes slower / decreases [1]

**(d)** **(i)** decomposition [1]

**(ii)** limewater [1]
**ALLOW**: calcium hydroxide solution

turns milky / cloudy / white ppt [1]
2nd mark dependent on first being correct

**(e)** **(i)** calcium nitrate [1]
water [1]

**(ii)** neutralise acidic soils / neutralise acidic lakes / making mortar / making calcium hydroxide / making limewater / whitewash [1]
**ALLOW:** making cement / making lines on roads (or games pitches) / in steel making

**(iii)** exothermic [1]

**[Total: 15]**

**4** **(a)** Bunsen burner / source of heat [1]
**ALLOW**: heating / heat

**(b)** X at 'space' at top of test tube [1]

**(c)** speed up the reaction / increase rate of reaction / make reaction go faster [1]

**(d)** $C_4H_8$ / $2C_2H_4$ [1]


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**(e)** **(i)** decolourises / goes colourless [1]
**IGNORE**: goes clear

**(ii)** B [1]

**(iii)** 4th box ticked (polymerisation) [1]

**(f)** **(i)** $C_7H_{16}$ [1]

**(ii)** substance containing carbon and hydrogen only [1]

**(g)** carbon dioxide [1]
water [1]
**ALLOW**: correct formulae

**[Total: 11]**

**5** **(a)** 3rd and 5th boxes ticked (sugar and water) (1 mark each) [2]

**(b)** (fractional) distillation [1]
**IGNORE**: fractionation

**(c)** O-H at right [1]
**ALLOW**: OH

**(d)** octanol [1]

**(e)** from ethene and steam [1]
**ALLOW**: from ethene and water
AND any two of: [2]
high temperature / heat / stated temperature between 150 and 1000 °C
catalyst / phosphoric acid
high pressure / stated pressure between 50-100 atm
**NOTE**: allow sulfuric acid (1) dilute with water (1) heat (1)

**[Total: 8]**

**6** **(a)** **(i)** reversible reaction / equilibrium reaction / reaction can go both ways [1]
**IGNORE**: products go to reactants / it is a reverse reaction

**(ii)** add water to white copper sulfate / add water to anhydrous copper sulfate
**ALLOW**: add water to $CuSO_4$ [1]

turns it blue [1]


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**(iii)** melt it / turn it to liquid [1]
dissolve it in water / make a solution of it [1]
**ALLOW**: add water

**(b) (i)** floats on top (of the mixture) / it is on top (of the mixture) [1]

**(ii)** S gains oxygen / it gains oxygen / S turns to $SO_2$ [1]
**ALLOW**: it / sulfur increases in oxidation number
**ALLOW**: it / sulfur loses electrons

**(iii)** cathode: C [1]
electrolyte: D [1]

**[Total: 9]**

**7 (a)** 112 ($^{o}$C) [1]

liquid [1]

**(b)** arrangement: go from regularly to irregularly arranged / become more irregularly arranged / go from regular to random [1]
**ALLOW**: idea of becoming less packed / less arranged / not so close together (but not implication of particles being apart from each other)
**NOTE**: do not allow implication of particles being 'apart' in solid
motion: start moving / start sliding over each other / go from no movement to movement / go from just vibrating to moving (over each other) [1]
**ALLOW**: idea of greater movement

**(c)** Any three of: [3]

(crystal) dissolves / idea of dissolving
particles (in crystal) become separated / solvent molecules get in between particles / mixing of particles / spreading out of particles
diffusion
movement of particles (in solution)
random (movement of particles)
particles collide
**ALLOW**: particles move from concentrated to dilute solution

**[Total: 7]**


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**8** **(a)** Any 2 of: [2]

compound has constant composition but mixture has variable composition
compound cannot be separated into different components (by physical means) but mixture can (be separated) / only the mixture can be separated
**ALLOW**: elements are chemically combined in compound but not in mixture
compound has properties different from elements it contains but mixture has the properties of the substances within it
**ALLOW**: compounds have sharp melting point (or boiling point) and mixture does not
energy change when compound formed but no (or very small) energy change when mixture formed

**(b)** Any two of: [2]

filtration / salts move to the clay pot and insoluble particles (remain) in the bowl
large particles (or insoluble particles) caught by leaves
the salts dissolve in the water / the salts are soluble
(dissolved) salts pass or through) the (holes in the) leaves /
**IGNORE**: salts pass through holes in the bowl

**(c)** **(i)** sodium carbonate [1]

**(ii)** chloride / $Cl^-$ [1]
**IGNORE**: chlorine

**(iii)** $K^+$ [1]
$SO_4^{2-}$ [1]

**(d)** 2 ($NaCl$) [1]

**(e)** electrons / an electron [1]
**IGNORE**: negative charge

**[Total: 10]**


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